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Dynamics of a policy-driven market: The co-evolution of technological innovation systems for solar photovoltaics in China and Germany

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ABSTRACT

The rise of Chinese solar energy firms has taken many experts by surprise. German policy makers and researchers alike had suggested that the country's ambitious deployment policies would translate into a competitive advantage for the German solar photovoltaics industry. This paper argues that these expectations rested on an outdated model of the international diffusion of innovation. Building on the technological innovation system (TIS) framework and the related system functions approach, the paper thus proposes a new approach for capturing the global dynamics of innovation and industrial development in emerging technology fields. Focusing on a period of dynamic growth in the field of crystalline-based PV technologies, the paper highlights how a set of dynamic and mutual inter-dependencies between an industrialized country (i.e. Germany) and an emerging economy (i.e. China) have driven the development and diffusion of technology in the field.

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1. Introduction

The dynamic development of the solar energy sector over the past decade has been closely linked to Germany's ambitious deployment policies. The German solar feed-in tariff triggered an unprecedented

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phase of market expansion and enabled the large scale mobilization of private investment for the production and installation of solar photovoltaic (PV) systems. From this perspective, Germany's solar energy policy has been a resounding success. The scale of deployment and the speed with which system prices have been reduced have vastly exceeded expectations (Hoppmann et al., 2014). Moreover, Germany's success inspired other countries to follow suit and introduce similar support schemes, enabling a global market for solar PV to take shape.

Simultaneously, important claims made by the proponents of Germany's ambitious deployment policy have not been fulfilled (BMBF, 2011). Market support was justified with the expected economic advantages for Germany (Bundesregierung, 2002). In the official justification of the revised Renewable Energy Act of 2008, the government cites the policy's role in generating so-called "first mover advantages" for German firms (Bundesregierung, 2008a). In a similar vein, Germany's environment ministry has formulated a strategy for "lead market" development via ambitious demand-side support for renewable energy technologies (Bundesregierung, 2008b). Following the reasoning of Jänicke and Jacob (2004), it argues that policy-induced lead markets for renewable energy technologies are accompanied by competitive advantages for domestic firms. Critics on the other hand claim that German deployment support has merely supported the build-up of a PV industry in China.[1]

As this paper will demonstrate, both lines of argument imply a linear concept of innovation and diffusion, which fails to capture the dynamic interaction between the two countries that has shaped the sector over the past two decades. The debate on lead markets draws on an outdated model of innovation and diffusion, which no longer adequately reflects the dynamics of an increasingly integrated, global economic system. Based on a simplistic lead-lag market model, its proponents have assumed that internationalization of demand drives a subsequent internationalization of supply (Beise, 2004; European Commission, 2006; Jänicke and Jacob, 2004; Quitzow et al., 2014). Following from this, firms in the lead market are expected to have an early mover advantage when new technologies diffuse globally. However, as China's rapid rise in the field of solar PV has revealed, this pattern no longer applies to important emerging technology fields. Rather than acting as mere adopters of technology, China and other follower countries have assumed active roles in shaping technology trajectories.

This paper proposes a novel framework to capture these emerging global dynamics of innovation and diffusion. To do so, it draws on the technological innovation system (TIS) literature and the related approach for analyzing TIS functional dynamics. Focusing on a period of dynamic growth in the field of crystalline-based PV technologies, the proposed framework is used to highlight how a set of dynamic and mutual inter-dependencies between an industrialized country (i.e. Germany) and an emerging economy (i.e. China) have driven development and diffusion of technology in the field. It demonstrates how uneven policy, market and industry dynamics in these spatially distinct entities have co-evolved and reinforced each other, thus enabling the dynamic market growth and industry expansion at the global level. This perspective serves as a complement to the contribution from Dewald & Fromhold-Eisebith in this special issue, which points out how processes at different yet overlapping spatial scales have influenced and mutually reinforced the development of Germany's TIS for solar photovoltaics (PV).

The paper begins with a review of the current debate on geographic dimensions in the analysis of TIS dynamics. Building on this, the paper proposes a novel analytical approach to capture geographical differences and cross-country inter-dependencies in the analysis of TIS functional dynamics. Next the empirical research case and methods of data collection are briefly presented. The analytical framework is then applied for the analysis of the TIS for solar PV in China and Germany. The paper closes with a synthesis of key findings and a number of policy-related conclusions.

2. Considering geographical dimensions of TIS dynamics

The system functions approach for analyzing TIS development, as proposed by Bergek et al. (2008) and Hekkert et al. (2007), has emerged as a useful framework for analyzing the dynamics of innovation

[1] See for example "The Sun Rises in the East: German Solar Firms Eclipsed by Chinese Rivals," *Spiegel Online*, September 07, 2011. Accessed on May 27, 2014 at http://www.spiegel.de/international/business/the-sun-rises-in-the-east-german-solar-firms-eclipsed-by-chinese-rivals-a-784653.html.

in emergent technology fields (Coenen et al., 2012; Coenen and Díaz López, 2010). Rather than focusing on static structures, i.e. a set of institutions, networks and actors, it focuses attention on how key processes in the system, or "system functions," evolve over time, allowing for related changes in the structural makeup of the TIS. Hekkert et al. (2007) have identified the following seven key functions, which jointly drive TIS development:

- "entrepreneurial activities," [F1]
- "knowledge development" [F2]
- "knowledge diffusion" [F3]
- "guidance of the search" [F4]
- "market formation" [F5]
- "resource mobilization" [F6]
- "creation of legitimacy" [F7]

The approach stresses that these system functions interact with each other in ways, which may positively or negatively affect overall system performance. The case of positive, mutually reinforcing functional dynamics is referred to as a process of "cumulative causation" (Bergek et al., 2008; Hekkert et al., 2007).

Although, in principle, a TIS spans geographic boundaries (Carlsson and Stankiewicz, 1991; Hekkert et al., 2007), in the past most studies have focused mainly on TIS development in industrialized countries, emphasizing dynamics at the national level. They have not considered cross-country interactions or other international influences in their treatment of a country-specific TIS (Binz et al., 2014; Coenen et al., 2012). In their study on Germany's TIS for solar PV cells, for instance, Jacobsson et al. (2004) focus on domestic functional dynamics, largely ignoring parallel developments in Japan or the US. Existing multi-country studies have largely taken a comparative approach without exploring the dynamic inter-linkages across countries (Gosens et al., 2013).

The failure to consider different spatial scales and their interactions has been identified as an important weakness in the TIS literature as well as the literature on sustainability transitions (Coenen et al., 2012). The studies in the field which do address geographical factors have mainly focused on the role of local, urban or regional context (Hansen and Coenen, 2013). Within the TIS literature, Dewald and Truffer's (2012) study on local processes of market formation in Germany's PV sector represent a case in point. They demonstrate the importance of dynamics at the community level in enabling the later success of Germany's national feed-in tariff in driving market development.

While such a localized view adds an important geographic dimension to the study of TIS dynamics, it needs to be complemented with an understanding of the complementarity and the interconnections between localized assets and institutions (Truffer and Coenen, 2012). In their relational perspective on TIS space, Binz et al. (2014) highlight the importance of networks at different spatial scales in shaping developments within national or regional sub-systems within a global TIS. Using social network analysis, they show empirically how international knowledge networks within the global TIS for membrane bioreactor (MBR) technology have developed and changed over time, suggesting that transnational linkages may play different roles at different stages of TIS development. They stop short of addressing how the transnational linkages and networks may influence TIS development, however.

To address this question, a number of studies have investigated how transnational linkages have influenced TIS development in emerging economies,[2] though largely adhering – albeit implicitly – to a lead-lag market model (Binz et al., 2012; Gosens and Lu, 2013; Schmidt and Dabur, 2014). Both Gosens and Lu (2013) and Schmidt and Dabur (2014) conceptualize TIS development in developing countries as a process of catching-up to a more advanced international TIS dominated by industrialized countries. Gosens and Lu (2013) acknowledge that the process of catching-up implies an evolution from a role of dependency to a role of contribution to the global TIS. However, both conceptually and empirically, they focus on the former. In their analysis of the TIS for MBR technology, Binz et al. (2012) go a step further by considering parallel dynamics in the broader international TIS and the Chinese sub-system.

[2] One exception is the comparative study of Vasseur et al. (2013) on the TIS for solar PV in Japan and Netherlands.

To enable a more nuanced analysis of the Chinese case, they then consider coupling patterns between the two levels. However, like the other studies, the analysis maintains the traditional assumption that technological development in the West influences dynamics in the developing world but not vice versa. While this may adequately represent the case of early phases of catching-up, the following case study will demonstrate that this does not do justice to the process of internationalization in the TIS for solar PV. Rather, the picture that emerges is one of increasingly interlinked TIS, where an industrialized (e.g. Germany) and an emerging country (e.g. China) have played different yet complementary and mutually supportive roles.

3. A conceptual approach to analyzing co-evolving country-level TIS

To capture the dynamic interaction between China and Germany in the PV sector, this paper proposes the concept of co-evolving, country-level TIS, characterized by "reciprocal influences" between the two co-evolving sub-systems (Murmann, 2003, p. 23). Following the typology of spatial TIS configurations proposed by Binz et al. (2014), this presupposes a global, multi-scalar TIS, where a number of internally cohesive sub-systems at the national (and sub-national) levels are inter-connected via transnational linkages. Furthermore, due to the predominantly national nature of deployment policies, it is assumed that country-level systems retain a degree of autonomy. Hence, national policy choices coupled with asymmetries in the respective national innovation systems are expected to yield differing functional dynamics, a finding also confirmed by comparative TIS studies (see Vasseur et al. (2013) for a PV-related example).

To facilitate the dynamic analysis of co-evolving, asymmetrical country-level TIS, it is suggested to combine two key analytical dimensions. The first dimension has the objective of identifying the asymmetrical functional patterns across the countries under consideration. To characterize evolving functional patterns over time, Suurs and Hekkert (2009) have proposed the "motor of innovation" concept. They suggest that different periods in TIS development are characterized by a unique confluence of system functions (i.e. the motor of innovation), which drive system building via dynamic processes of cumulative causation. The concept helps focus attention on the central drivers of development and change during a particular time period and how they interact to unleash processes of cumulative causation. The following analysis focuses attention not only on functional patterns across different time periods, but also across space (i.e. different countries). To highlight spatial asymmetries, the "motor of innovation" concept is adopted as a tool for characterizing key country-level system-building dynamics in the chosen time periods. After a description of system functions based on Hekkert et al. (2007) in both Germany and China, it is employed as a focusing device for contrasting the core system dynamics in the two countries.

Taken in isolation, such an analysis would represent a simple comparison of TIS functional patterns. This first analytical dimension is, therefore, supplemented by a second dimension, which captures relevant cross-country inter-dependencies and how they contribute to TIS dynamics in the respective countries. The mentioned studies on TIS development in developing countries have already pointed out that system functions in one country may influence TIS dynamics in another via so-called "transnational linkages" (Gosens and Lu, 2013) or "couplings" (Binz et al., 2014) (referred to as transnational linkages in the remainder of the paper). Drawing on different streams of literature, including the literature on technology transfer, global production networks and transnational entrepreneurship, these studies have highlighted mechanisms through which cross-country influences within the different system functions might take shape. They refer to direct linkages, as manifested, for instance, in the form of relationships within a global production network or transnational knowledge networks (Binz et al., 2014; Gosens and Lu, 2013). Among other things, these relationships may enable cross-country knowledge spillovers, thus representing a bridge between the processes of knowledge development and diffusion within the countries involved (Hoppmann et al., 2013). Similarly, resource mobilization may be supported via international financial markets and the underlying networks and institutions (Binz et al., 2014).

This paper proposes that cross-country influences may also take shape via more indirect channels, which occur independently from linkages among actors directly involved in the respective country-level TIS. In the following, two different forms of indirect influences are identified. Firstly, so-called

"indirect spillovers" may take shape, when developments in one country stimulate similar developments abroad without direct interaction between actors within the TIS. For instance, market growth in one country may spill over into other countries as a result of a "demonstration effect," as referred to in the literature on the diffusion of innovations (Beise, 2004). By reducing uncertainty regarding a given technology, market formation in one country may stimulate or reinforce market development in other countries without direct linkages being necessary. Similarly, the policy diffusion literature has pointed out that the adoption of policies by other countries may increase domestic pressures to adopt similar policies without any "agents" of policy transfer being involved (Berry and Berry, 1990). In a similar vein, Bergek et al. (2008) have noted that important TIS developments may have an influence on the direction of search beyond the country where they are taking place.

In addition, indirect cross-country influences may follow less linear patterns, taking the form of more complex "feedbacks." For instance, competitive pressure may trigger policy action aimed at distinguishing the domestic policy environment from policy practices in other countries (Tews, 2005). Moreover, highly visible events in a country may stimulate other types of responses in foreign countries without any directly related interaction between the countries taking place. The final decision to phase out nuclear energy in Germany after the Fukushima accident is a case in point (Wittneben, 2012). While it should be acknowledged that the described indirect spillovers and feedbacks remain dependent on more generalized cross-country linkages (e.g. in the form of a globalized mass media networks) these linkages are considered external to the TIS.

4. Research case, methods and data

Based on the conceptual framework outlined above, Section 5 presents empirical data on the co-evolutionary dynamics of TIS development in the solar PV sector from 1999/2000 to 2010/2011, the most dynamic growth phase in the sector to date.[3] The analysis focuses primarily on the dynamic interactions between Germany and China, as these have represented the main driving forces of global TIS development during the period. Ideally, the analysis would also include an in-depth treatment of other key countries. Since this is beyond the scope of the paper, it focuses on Germany and China, considering important developments in third countries only where essential. Although this cannot do justice to the full complexity of global TIS development, it provides a sufficient empirical basis for understanding the core dynamics of growth and internalization in the TIS for solar PV and testing the suggested framework for analyzing the co-evolutionary dynamics of TIS development.

The analysis is divided into three time periods, each marked by important turning points in TIS development. For each period, the paper describes the uneven functional dynamics between the two countries and highlights key cross-country linkages as well as more indirect spillovers and feedbacks. By sub-dividing the analysis in this way, the paper emphasizes the evolving nature of TIS functional dynamics and how the respective roles of the countries have changed over time. It shows how country-level dynamics have affected overall TIS development, including important geographical shifts.

It begins with the time period from 1999/2000 to 2003. This reflects the lifetime of Germany's 100,000 rooftop programme and represents an important transitional period in the PV market. Firstly, the centre of demand began shifting from Japan to Germany, and secondly, the market shifted from the previous dominance of off-grid applications to an overwhelming dominance of policy-driven, grid-connected systems (Mints, 2012). Due to space constraints, this transitional period is only briefly summarized and mainly serves as a background to the more detailed analysis of the remaining period.

The period from 2004 to 2010/2011 is considered the core growth period and is further subdivided into the periods from 2004 to 2008 and 2009 to 2010/2011. The former begins with the introduction of Germany's federal, cost-covering feed-in tariff and the removal of the previous cap on policy-supported market growth. It is the first stage of rapid growth in Germany's policy-driven market, albeit constrained by rising module prices. Finally, the period from 2009 to 2010/2011 represents the steepest jump in annual installed capacity and was characterized by a drastic decline in module prices. While

[3] Annual installed capacity increased by a factor of 69 over the period, compared to a factor of 13 and 6 in the periods 1980–1990 and 1990–2000, respectively.

the global market has continued to grow thereafter, market growth in Germany and other European countries has declined, representing another important turning point in TIS development.

The empirical data that are presented constitute the results of 35 interviews conducted mainly in Germany and China with representatives from PV manufacturing firms, policy officials as well as industry experts from consulting firms, industry associations, research institutions and international organizations (such as IEA PVPS).[4] To triangulate findings from the interviews, this was followed up by a review of key indicators available in relevant primary sources, including annual reports of listed PV firms and policy documents, and secondary sources, including articles published in specialized media outlets and academic journals. Finally, to avoid factual errors, an extended version of this article was reviewed by two industry experts in China and Germany.

5. The co-evolution of uneven functional dynamics in China and Germany and the role of cross-country influences

The following section provides an analysis of TIS functional dynamics in Germany and China for the three periods indicated above. Each section closes with a synthesis, including a summary of results in form of a table and a graphical illustration of transnational linkages. The purpose of the synthesis is to identify key motors of innovation in the two countries and to highlight the role of transnational linkages, spillovers and feedbacks.

5.1. The preliminary growth phase (1999–2003): 100,000 rooftops and Chinese manufacturing pioneers

5.1.1. Summary of functional dynamics in Germany

As alluded to above, the period from 1999 to 2003 represented a transitional period in the PV sector, in which Germany established itself as the main driver of global demand. This was closely related to the election of the first federal government with participation from the Green Party in 1998 (Fuchs and Wassermann, 2012). This led to the concomitant shift from a market driven by multiple local and regional initiatives to a national market driven by the federal government's 100,000-rooftop programme (Dewald and Truffer, 2012). In tandem with the national feed-in tariff, the programme offered concessional financing for a total of 350 MW of mainly privately owned rooftop systems, reaching this target in 2003 (Bruns et al., 2011, pp.197–199; KfW, 2004, p.45).

German market development also enabled the growth of a number of German solar manufacturing start-ups, and by 2003 most of today's major German PV equipment suppliers had entered the market. With this Germany had established a presence along the entire PV value chain. Nevertheless, during the period, Japan still accounted for over 50% of global production of cells and modules, and more than half of Germany's demand was being met by foreign-produced products (KfW, 2004, p.49). Similarly, Germany had developed substantial R&D capacity by the early 2000s, though lagging behind Japan and the US in terms of R&D spending (Räuber, 2005, p.152).

5.1.2. Summary of functional dynamics in China

As Germany began emerging as the world's leading PV market, modest market developments in China were driven by a number of rural electrification programmes (NREL, 2004a, 2004b). Simultaneously, the production of solar modules and cells began to increase slowly from the low levels of the 1990s, reaching an estimated 35 MW in 2003 (Li, 2004, p.78). More importantly, the company mix began to shift from state-owned firms focused exclusively on the small domestic market to privately held enterprises. These included Trina, Yingli, Suntech and Canadian Solar, which entered the market in the late 1990s and early 2000s (Marigo, 2007, p. 146). While these new companies profited from early sales in the small domestic market, their commercial orientation soon led them to target the booming European market (Zhang and White, 2012, p. 36).

[4] See the Appendix for an overview of interview partners in China and Germany.

The establishment of these pioneer firms played a pivotal role in preparing the ground for the dynamic developments of the Chinese PV industry in the subsequent periods. As Zhang and White (2012) have argued, these early movers played a critical part in overcoming what they call the industry's "liability of newness" in China. They established firms in the solar PV sector as recognized high-technology enterprises and demonstrated the ability to compete with more established international players (Zhang and White, 2012, p. 35). Moreover, they became important vehicles for the acquisition of cutting-edge foreign expertise via the return of experienced, foreign-trained Chinese professionals from North America and Australia (including the CEOs of Suntech and Canadian Solar) (De la Tour et al., 2011, p. 764–765). Under the able leadership of these foreign-trained executives, the firms were able to assemble production lines with imported machinery from equipment suppliers in Europe and the US.

5.1.3. Synthesis: motors of innovation and cross-country inter-dependencies

German functional dynamics during this period were largely driven by the fortuitous political developments at the federal level. This allowed the experience gathered by numerous local initiatives in the 1990s to be translated into a strong federal market support scheme (Mautz and Byzio, 2005, p.79). The motor of innovation for the period is, therefore, labelled "Scaling-Up of Deployment Policies" and highlights the dynamic of mutual reinforcement between the supportive political climate and the acceleration of market growth (see also Dewald & Fromhold-Eisebith, 2015). Firm entries and investments in manufacturing capacity and R&D followed, but they could not keep pace with the dynamic demand-side developments. As a result, the German market in this period remained heavily dependent on linkages with foreign producers, especially in Japan and the US, who supplied the majority of the market [Transnational Linkage #1, TL1].

In China, the motor of innovation is simply labelled "Entrepreneurial Activities," the clearly dominant driver of system building in this period. Moreover, the entry of the first privately owned PV firms in China went hand in hand with a stronger outward orientation of the Chinese PV sector. Simultaneously, the emerging gap between demand and (domestic) supply in Germany represented an important opportunity for less established producers to enter a sector previously dominated by Japanese market demand and its strong domestic suppliers. Before any direct cross-country linkages emerged, this market opportunity in Germany offered important "guidance of the search" for Chinese entrepreneurs who began pursuing strategies aimed at establishing commercial PV firms [Indirect Spillover #1, IS1]. In some cases, it inspired foreign-based – or "transnational" (Drori et al., 2009) – entrepreneurs to return to China, establishing linkages with foreign centres of knowledge creation [TL2]. Moreover, with German equipment suppliers entering the PV sector, production technologies, previously developed mainly in-house by Japan's large producers, became more readily available on the global market, enabling first linkages between Chinese manufacturers and German equipment providers to emerge [TL3] (Table 1) (Fig. 1).

Table 1
Motors of innovations in Germany and China and cross-country inter-dependencies (1999–2003).

Motors of innovation	
Germany	China
Scaling-up of deployment policies	Entrepreneurial activities
Transnational linkages	
• Linkages between Japanese and American module producers and German distributors and project developers [TL1] • Emerging process of knowledge transfer facilitated by Chinese returnees especially from Australia and North America [TL2] • Emerging linkages between Chinese pioneer firms and German and other international equipment suppliers [TL3]	
Indirect spillovers and feedbacks	
• Indirect Spillover [IS1]: German market developments provide *guidance of the search* in China	

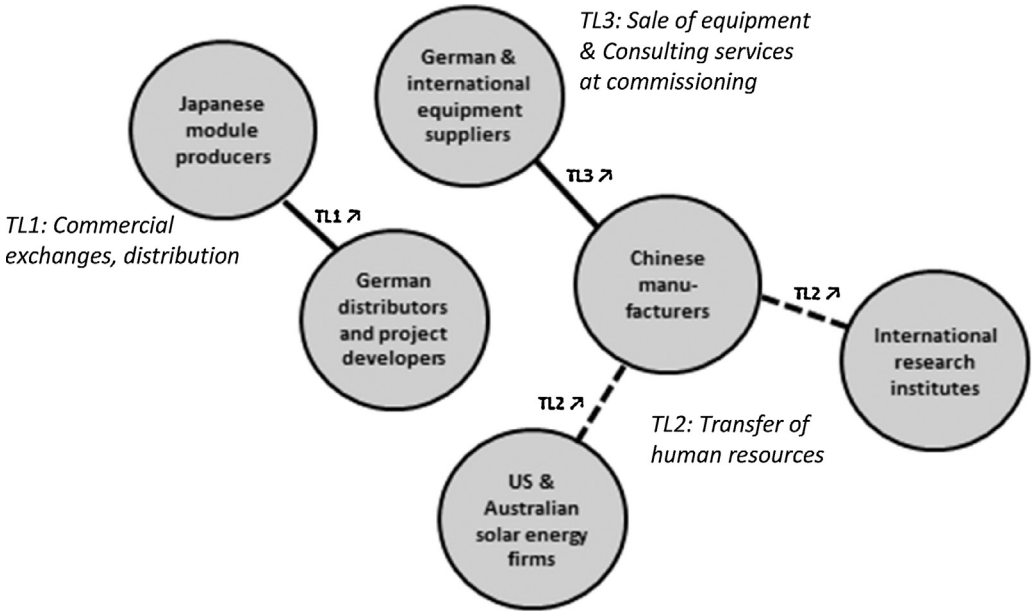


Fig. 1. Emerging transnational linkages (1999–2003) (*This and subsequent diagrams do not present *all* transnational linkages but focuses on those linkages, which are considered particularly important for TIS development in Germany and China.).

5.2. *The first growth phase (2004–2008): strong market growth despite increasing module prices*

5.2.1. *Key functional dynamics in Germany*

For Germany, the period from 2004 to 2008 represented a period of growth and consolidation of demand-side support. Growing employment in the industry further bolstered the sector's political clout, so that sustaining the emergent solar industry played a central role in justifying the increase of the solar feed-in tariff in late 2003 (Deutscher Bundestag, 2003, p. 1) [F7]. The increased feed-in tariff made investments in PV systems profitable without supplemental capital subsidies, so that market growth was no longer constrained by a cap on programme funds. This increased investment security and helped to further accelerate market growth in 2004 and 2005. In 2006, the market slowed down slightly, due to supply shortages and rising module prices. However, by 2007, these price increases could be compensated by rising efficiencies in the balance of systems (i.e. the integration of the different components in a PV system). This enabled overall system prices to decline, so that the market grew by a further 50% in 2007 and 2008 (see Fig. 2) [F5].

"Resource mobilization" accelerated in tandem with market development albeit with a focus on the demand-side. Investments in PV systems rose progressively from €1,5 billion in 2004 to an annual expenditure of €6,2 billion by 2008. Close to half of demand-side investments benefited from low-interest loans supplied by Germany's development bank (Kreditanstalt für Wiederaufbau, KfW). In addition, local banks became important providers of capital for the large number of privately owned systems, while equity was still provided mainly by individuals and farmers (Lütter et al., 2009, p. 140). For the growing number of larger systems, project-based financing was largely provided by regional State-owned banks (*Landesbanken*) as well as the Commerzbank, which had gained significant experience in the wind sector (Quitzow, 2013, p.24). German supply-side investments also increased but neither kept pace with German market growth nor with global investment trends. Despite the availability of public investment subsidies, especially in East Germany (Brachert et al., 2013; Grau et al., 2012), the share of global supply-side investment decreased from a high of 90% in 2002 to approximately 15% in 2007 (Jennings et al., 2008). Consequently, domestic firms consistently supplied less than 50% of German demand (Lütter et al., 2009) [F6].

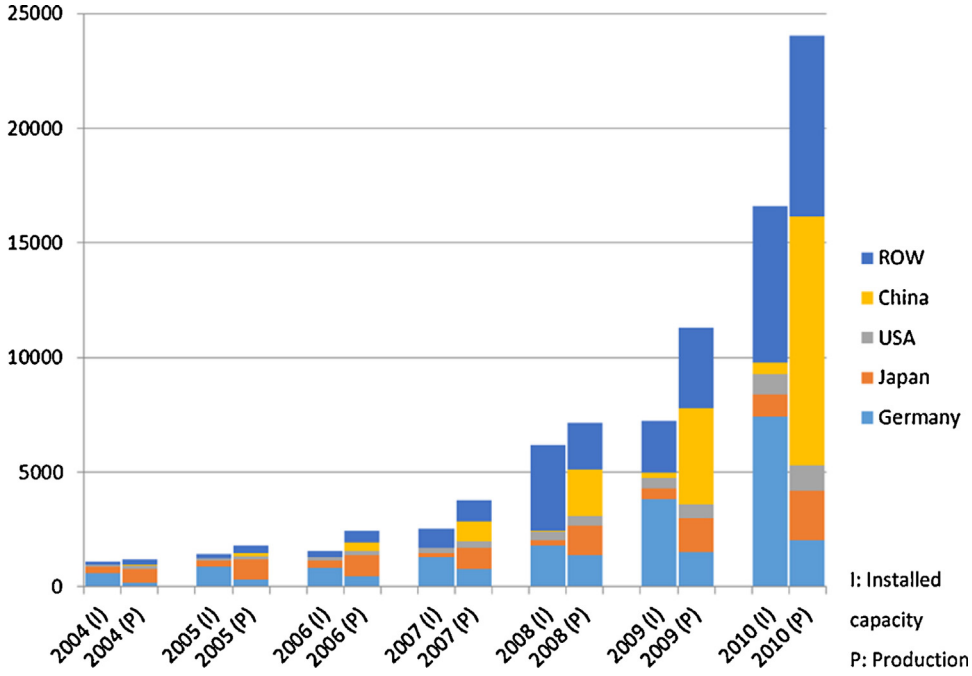


Fig. 2. Development of installed capacity and solar PV production in selected countries (2004–2010).
Source: Earth Policy Institute Database

As alluded to above, German industry proved relatively ineffective at reducing prices for cells and modules, which actually increased in this period (Mints, 2012). In particular, the supply of silicon did not keep pace with market developments, resulting in steep price increases (US Department of Energy, 2011, pp.33–35; Willeke and Räuber, 2012, p. 36). This not only increased input costs, but it also affected the willingness of German cell and module producers to expand production. As Germany's largest cell producer, Q-Cells, states in its 2005 annual report, "Our policy is to only expand our production capacity when we have access to the corresponding raw materials. . ." (Q-Cells, 2006, p.21) [F6].

Q-Cells along with its main rival SolarWorld represented Germany's most important drivers of "entrepreneurial activities" during the period. Q-Cells with its focus on cell manufacturing became the world's largest supplier of solar cells in 2007, while Solar World integrated vertically via multiple acquisitions. In addition, a number of equipment suppliers entered the PV sector, while existing suppliers expanded their PV activities [F1]. The equipment sector demonstrated the highest annual investment ratio, ranging from 14 to 25% (Lütter et al., 2009) [F6].

"Knowledge development" in Germany benefited from the sector's increased political standing with government expenditures almost doubling in 2004 and remaining stable thereafter (BMU, 2011, p.11). Spending focused on traditional silicon-based solar cells with slightly over 50% of funding as well as thin film technologies with approximately 30% of funding (Quitzow, 2013, pp.29–30). Private R&D investments also increased in tandem with industry turnover. In 2006, private R&D investments reached €65 million, surpassing public expenditures for the first time. By 2008, private expenditures had reached more than five times the public budget (BSW and Solarpraxis, 2009, p.31) [F2]. A particular strength of the German R&D landscape has been the close collaboration between manufacturers and equipment suppliers and dense network connections, enabling rapid knowledge diffusion within the sector (Hornych and Brachert, 2010; Nussbaumer et al., 2007) [F3].

As in the past, the biannual "Glottertaler Gespräche" represented an important forum for industry stakeholders and policy makers to discuss the strategic direction of R&D programmes

(Bruns et al., 2011, p.172). According to the research agenda developed in 2005, twin goals of public R&D funding were to enhance the competitiveness of solar energy vis-à-vis conventional energy and secure Germany's technological leadership in the PV sector (BMU, 2006, p.12). Additional "guidance of the search" resulted from the emerging silicon shortage, driving R&D aimed at reducing the amount of silicon needed in the production process. Furthermore, it stimulated investments in the development of thin film technologies, which require only limited or no silicon (Quitzow, 2013) [F4] (For a summary of functional dynamics, see Table 2).

Table 2
Summary of functional patterns in Germany and China (2004–2008).

	Germany	China
F1: Entrepreneurial activities	Few entries in manufacturing and modest expansion of existing firms led by two market leaders; Entries and expansion in equipment supply	Large-scale entry and rapid expansion of production capacity
F2: Knowledge development	Strong growth in private R&D funding and important R&D collaborations between manufactures and equipment suppliers	Low levels of R&D focused on silicon production; Learning by doing focused on cost-reducing process innovation; Integration of foreign and domestic knowledge resources
F3: Knowledge diffusion	Diffusion of knowledge via dense network connections in the industry and intense collaboration between manufacturers and equipment suppliers	Acquisition of international manufacturing know-how via equipment purchases, international expert migration and collaborations with German manufacturers and international R&D centres
F4: Guidance of the search	Public–private dialogue on R&D agenda and market pressures to reduce the use of silicon inputs	Public research agenda focused on attaining technological capacities to produce silicon; Market requirements in Germany and other European countries, such as quality standards
F5: (Domestic) market formation	Revision of solar feed-in tariff enables strong growth in domestic market	Low but stable growth of domestic market
F6: Resource mobilization	Strong growth of demand-side investment, moderate growth of supply-side investments; decrease of global investment shares	Strong supply-side investment growth and mobilization of foreign public equity to fund industry expansion; increase in global investment shares
F7: Creation of legitimacy	Strengthened solar industry provides the sector with increased political weight at the federal level	Growing international legitimacy of Chinese solar PV firms enabled by supply shortages in Germany

5.2.2. Key functional dynamics in China

In China, market growth until 2008 remained minimal relative to the rapidly expanding markets in Europe [F5]. Therefore, all major producers generated the vast majority of their sales overseas, most importantly in Germany (Quitzow, 2013, p. 34). Starting in 2006, the Spanish market became another important destination for Chinese module exports, peaking in 2008 (Kirkegaard et al., 2010, p.54). While some broader advances were made in the domestic legitimation of renewable energy in China, the international market success of China's module manufacturers provided the most important boost to the country's solar energy sector. An increasing mismatch between supply and demand in Germany (see Fig. 2) proved to be the key driver in helping China's pioneer firms to overcome important challenges related to quality concerns and the lack of a domestic certification infrastructure. Faced with supply constraints in Germany, leading German PV firms entered supply, distribution and OEM agreements with the Chinese pioneer firms. The strong credibility of these German companies played a crucial role in overcoming initial customer hesitation and establishing China's PV manufacturers as legitimate competitors in the German market (Hug and Schachinger, 2006) [F7].

These success stories stimulated aggressive investment strategies by the pioneer firms. Building on China's enabling environment for low-cost, high volume manufacturing, they combined rapid capacity expansion with varying degrees of vertical integration. Additionally, a second wave of major Chinese entrants began their operations progressively upstream in more capital-intensive segments of the value chain (Gallagher and Zhang, 2013; Zhang and White, 2012). These strategies were partly enabled by the parallel emergence of a large number of second- and third tier module manufacturers, some of which were acquired by the larger players. The aggregate result was a rapid increase in low-cost manufacturing capacity, representing more than a quarter of global capacity by 2008. Complementing this, a growing domestic network of suppliers to the PV industry began to develop (Gallagher and Zhang, 2013)[F1].

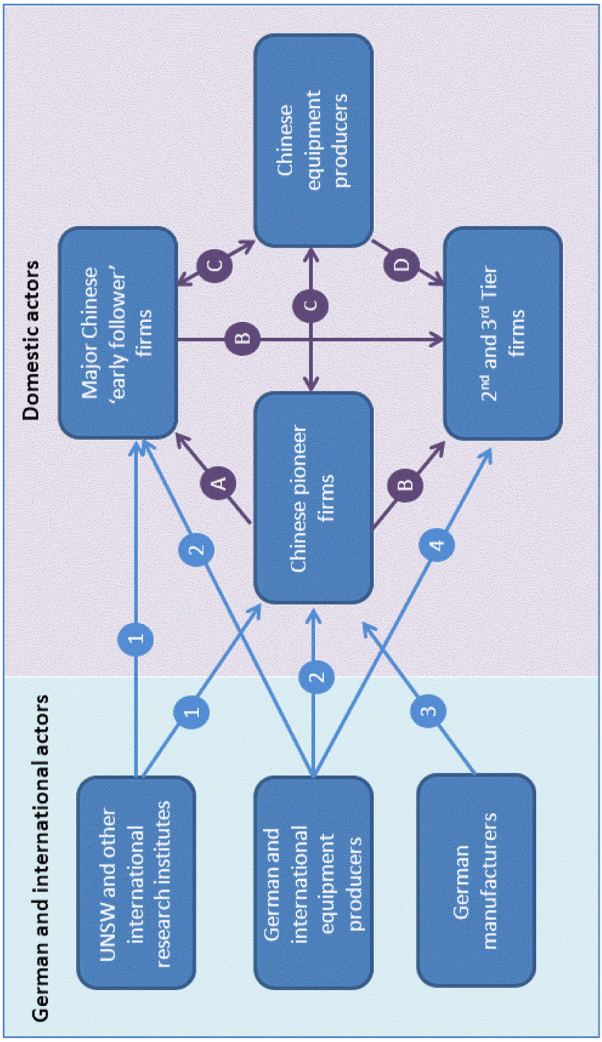
These investment strategies were facilitated by a transition from domestic finance, including China's "big four" State-owned banks, to the development of linkages with international private equity investors. The latter ranged from smaller VC and equity funds to major international investment banks (Zhang and White, 2012, pp.52–53). These financing partnerships prepared the ground for a wave of IPOs in New York, representing over 50% of global public equity activity in the sector between 2005 and 2008 (Quitzow, 2013). Complementing these commercial sources of finance, Chinese firms could also rely on a number of subsidy and investment support schemes at the local and provincial level. Local governments in China not only help arrange financing but typically also offer hi-tech companies tax breaks, access to low-cost energy and land and different forms of investment grants (Haley and Haley, 2013). Most of the major PV manufacturers note the receipt of investment or R&D grants in their annual reports starting in 2005 or 2006 (Quitzow, 2013)[F6].

"Knowledge development," on the other hand, still lagged significantly behind the leading countries of Germany, Japan or the US (Quitzow, 2013, p.44). Public R&D funding increased to a modest six million US dollars per year for the period 2006–2010 (Huo and Zhang, 2012) and was aimed in particular at the field of silicon production, in which patenting increased substantially (De la Tour et al., 2011; Fischer, 2012). PV firms, on the other hand, were focused mainly on acquiring existing know-how to launch state-of-the-art production lines, while reducing costs via process optimization (Quitzow, 2013, p.42–43)[F3]. Fig. 3 below provides a stylized depiction of the key international and domestic mechanisms of "knowledge diffusion." The most important linkages remained the relationships between German equipment suppliers and Chinese manufacturing firms. With a world market share of 40–50%, German equipment producers supplied the largest share of foreign machinery to Chinese firms (Wessendorf, 2012). Moreover, the sale of equipment was typically accompanied by consulting services for commissioning. Finally, from 2006, so-called turnkey production lines appeared on the market. These were purchased mainly by 2nd and 3rd tier producers in Asia without the in-house expertise available at larger firms.[5]

Further linkages emerged between leading German manufacturers and a number of earlier Chinese entrants. To meet the burgeoning demand in Germany, a number of manufacturers had explored the possibility of establishing OEM agreements and joint ventures with Chinese manufactures. In this context, they supported these firms in meeting international quality standards in cooperation with relevant certification bodies (Quitzow, 2013). Such exchanges were complemented by a continued migration of experts from Australia and North America (De la Tour et al., 2011, p.765). In particular, the School of Photovoltaic and Renewable Engineering at the University of New South Wales (UNSW) in Sydney became a key source of expertise, supplying a number of Chinese firms with key personnel (Zhang and White, 2012).

While knowledge flowed mainly from Europe, the US and Australia into China in this period, it is important to note that sources of knowledge were multiple and were being combined with advanced Chinese knowledge in mass production, acquired in the electronics sector and other adjacent industries (Zhang and White, 2012). This suggests that the stock of knowledge being developed in China cannot simply be equated to pre-existing knowledge resources in the West. Rather a unique Chinese knowledge base emerged (see Berkhout et al. (2011) for a discussion of this phenomenon), facilitated by transnational entrepreneurs who acted as key integrators of knowledge resources. In a subsequent

[5] Based on interviews with industry experts.



International knowledge transfer mechanisms

1. Transfer of personnel, R&D cooperation, licensing agreements; 2. Equipment purchase, consulting services and transfer of personnel; 3. Coaching; 4. Purchase of (turnkey) equipment and consulting services;

Domestic knowledge transfer mechanisms

A. Transfer of personnel; B. Cooperation via OEM agreements and cross-licensing; C. Mutual learning via equipment purchase and installation; D. Equipment purchase and installation

Fig. 3. International and domestic mechanisms of knowledge diffusion in China.

step, this new knowledge base diffused domestically via the transfer of personnel from China's pioneers to the early followers [F3].

Lacking a strong public R&D programme, Chinese firms were guided by a combination of market opportunities in Germany (and other European markets) and the country's strong political framework focused on promoting export-oriented, hi-tech industries (Petti, 2012). The key goal was to reach European and US quality standards, while attaining cost leadership (Mints, 2012) [F5] (For a summary of key functional dynamics, see Table 2).

5.2.3. Synthesis: motors of innovation and cross-country inter-dependencies

In Germany, this phase was characterized by strong market growth and corresponding mobilization of demand-side investments. This was enabled by the further institutionalization of Germany's market support scheme and the corresponding downstream market infrastructure, including project development, installation services and financing. The motor of innovation for the period is thus labelled "Institutionalization and Scaling-up of the Market." This was matched by a corresponding "Institutionalization and Scaling-up of Low-Cost Manufacturing" in China. Building on the exploratory work of the pioneers and China's overall strength in low-cost manufacturing, both first movers and fast followers were not only able to mobilize local government support but also large volumes of international private and public equity.

As a result, supply bottlenecks in Germany were increasingly being filled with lower cost modules from the rapidly expanding Chinese producers. These reached German markets via distribution and OEM agreements with leading German firms [TL4]. Simultaneously, Chinese industrial development was facilitated not only by the profits generated in Germany, but by important inflows of knowledge and finance. German equipment suppliers played a central role in transferring state of the art equipment and related know how, while their Chinese customers provided the needed demand to sustain the high investment ratios in the sub-sector [TL3]. This was complemented by continued migration of experts from Australia and North America [TL2], emerging knowledge partnerships with international R&D centres [TL5] as well as informal coaching by German manufacturers [TL6] and cooperation with certification bodies [TL7]. Financial inflows had multiple origins, but were dominated by the IPOs in New York [TL8]. Less tangible but equally important for Chinese industrial development was the positive feedback from market success in Germany on the domestic legitimation of the industry [Feedback #1, FB1]. Among other things, this enabled the needed local-level political support, a key to firm success in the Chinese context (Table 3) (Fig. 4).

Table 3
Motors of innovation in Germany and China and cross-country inter-dependencies (2004–2008).

Motors of innovation	
Germany	China
Institutionalization and scaling-up of the market	Institutionalization and scaling-up of low-cost manufacturing
Transnational linkages	
• Relative decline in importance of linkages between Japanese and American module producers and German distributors and project developers [TL1] • Increasing process of knowledge transfer facilitated by Chinese returnees, especially from Australia and North America [TL2] • Increasing linkages between Chinese pioneer firms and German and other international equipment suppliers [TL3] • Commercial linkages between pioneer firms and German manufacturers, distributors and project developers via supply, distribution and OEM agreements [TL4] • Emerging knowledge partnerships of between Chinese firms and international R&D centres [TL5] • Informal coaching of certain Chinese pioneers by German manufacturers [TL6] • Cooperation with German certification bodies to attain certification of quality standards [TL7] • Increasing linkages between Chinese firms and international capital markets [TL8]	
Indirect spillovers and feedbacks	
Feedback: Export performance helps establish credibility of Chinese firms first internationally and then in China [FB1]	

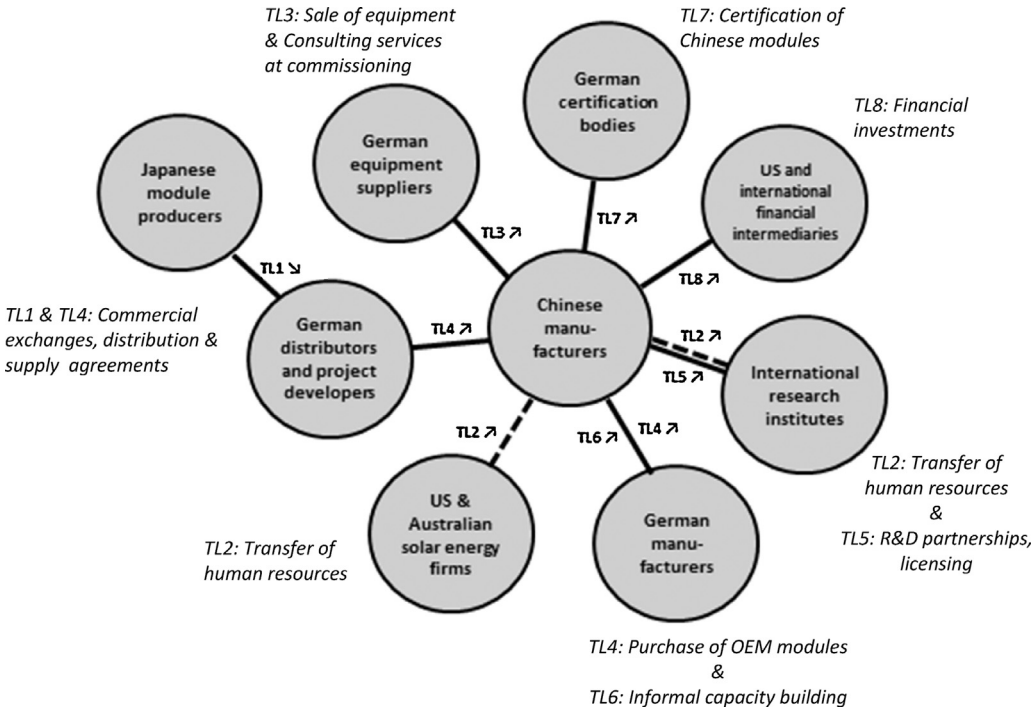


Fig. 4. Growth and adjustment of transnational linkages (2004–2008).

5.3. The second growth phase (2009–2010/2011): falling module prices trigger exponential market growth and policy feedbacks in Germany and China

5.3.1. Key functional dynamics in Germany

The period from 2009 to 2010/2011 was marked by the end of the global silicon shortage, triggering a sudden over-supply of modules as producers around the world ramped up production to their full capacity. The resulting decline in module prices vastly outpaced annual reductions in feed-in tariffs, prompting a precipitous increase in market growth in Germany [F5]. Simultaneously, the market share of German producers began to decline (see Fig. 5), raising concerns among national policy makers regarding the aggregate costs and benefits of the feed-in tariff. These trends were counter-balanced, however, by the growing political clout of the solar industry at the regional level, in particular in East Germany [F7]. As a result, more drastic amendments to the Renewable Energy Law proposed by the federal government were prevented by *Länder* governments in the *Bundesrat* (Germany's Upper Chamber) and market growth could continue unabated through 2011 (Bauchmüller, 2012; Bundesrat, 2010) [F5].

As in previous periods, dynamic market growth in Germany went hand in hand with record investments in the installation of solar systems. Despite the financial crisis, investments reached €9,6 billion in 2009 and then peaked at €19,5 billion in 2010. Moreover, financing became further diversified. From 2008, both solar energy cooperatives and closed solar funds[6] emerged as important additional sources of private capital for larger solar projects (Holstenkamp and Ulbrich, 2010; Quitzow, 2013). Meanwhile, in the wake of the financial crisis, supply-side investment showed important signs of weakness. In 2009, investments dropped to less than a quarter of the 2008 level[7] [F6]. Similarly,

[6] Closed solar funds represent a vehicle for mobilizing private equity for a determined set of solar energy projects.
[7] Based on global investment data from Bloomberg New Energy Finance Investment Database.

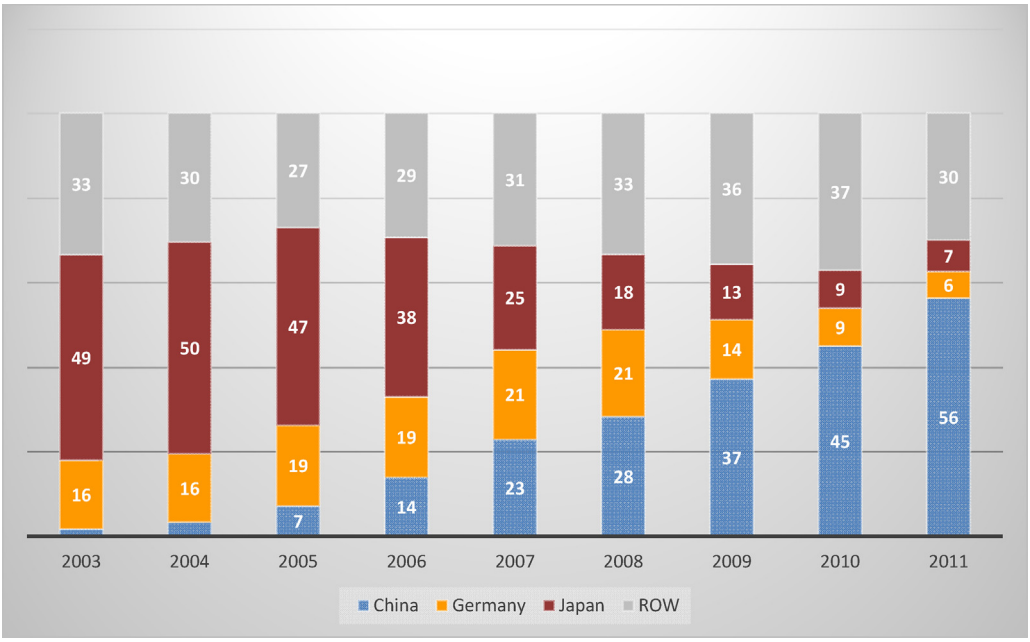


Fig. 5. Global share of production in China, Germany and Japan from 2003 to 2011.

"entrepreneurial activities" declined. The only major entry to the German PV industry during this period was Bosch, which entered the sector in 2008 with a range of acquisitions along the value chain (Quitzow, 2013, p. 21–23). As a result, Germany's share of global module production had dropped to less than 10% in 2010 (see Fig. 5) [F1].

Public sector R&D funding, on the other hand, increased to an average of €37 million in new spending between 2008 and 2010 [F2]. In addition, the new cluster initiative "Solar Valley Mitteldeutschland" was founded and the "Innovationsallianz Photovoltaik" was created to provide a further boost to cooperative R&D efforts [F3]. R&D intensity among the leading German manufacturers also remained high relative to industry standards. Moreover, fuelled by revenue from their international clients, German equipment suppliers remained at the forefront of technology development, exhibiting some of the highest R&D spending in the sector (Breyer et al., 2010, p.9–10) [F2]. The direction of public R&D funding was upheld, maintaining a balance between silicon-based cell concepts and thin film and other alternative technologies (BMU, 2010) [F4] (For a summary of key functional dynamics, see Table 4).

5.3.2. *Key functional dynamics in China*

As in the previous period, Germany's vigorous market expansion was met by continued expansion of manufacturing capacity by both new and existing Chinese firms. This allowed China to further increase its share of global production, reaching 45% in 2010 and over 50% in 2011 [F1]. Moreover, the increasing inroads of Chinese firms in the German market had boosted their reputation both internationally and in China. This translated into increasing political support first at the provincial and finally at the central level (Quitzow, 2013). In 2010, China's State Council proclaimed the solar PV industry as a "strategic emerging industry" (Ahrens, 2013, p. 5; Zhang and White, 2012, p. 3) [F7].

This central government decision coincided with a series of multi-billion dollar loan agreements between China Development Bank (CDB) and five major PV manufacturers.[8] Due to their relatively

[8] Mercom Capital, "Loans and Credit Agreements involving Chinese Banks to Chinese Solar Companies since January 2010," September 28, 2011. Accessed on October 27, 2013 at http://mercomcapital.com/loans-and-credit-agreements-involving-chinese-banks-to-chinese-solar-companies-since-jan-2010.

Table 4
Summary of functional patterns in Germany and China (2009–2010/2011).

	Germany	China
F1: Entrepreneurial activities	Entry of Bosch and further expansion of Q-Cells and SolarWorld	Strong expansion of production among existing firms and several significant entries
F2: Knowledge development	Continued investment in R&D by private and public actors, in particular in equipment sector	Increased domestic knowledge development in collaboration with international research centres
F3: Knowledge diffusion	Increased focus on cluster development and collaborative R&D; Purchase of existing firms by Bosch facilitates corresponding acquisition of knowledge	Continued acquisition of knowledge via equipment purchases, expert migration, licensing, etc.
F4: Guidance of search	Priorities of public–private dialogue remain focused on a balance between silicon-based cells and alternative technologies, in particular thin film technologies	Continued focus on process innovation aimed at cost reduction and development of more efficient cell concepts with guidance from international research institutes
F5: (Domestic) market formation	Falling module prices trigger exponential market growth in demand	Uncertainty in Europe coupled with growing domestic industry trigger first significant market support programmes
F6: Resource mobilization	Strong growth of demand-side investment, shrinking supply-side investment lead to further decrease of global investment shares	Continued strong investment growth and mobilization of foreign public equity to fund industry expansion, enabled by loan agreements with China Development Bank
F7: Creation of legitimacy	Signs of weakness at the federal level are counter-balanced by growing political clout at the *Länder* level	Growth and export performance of Chinese manufacturers provides significant boost to the sector's political importance

high interest rates, CDB credit lines are typically not drawn down by the firms. However, they function as *de facto* repayment guarantees for commercial lenders (Sanderson and Forsythe, 2013). This enabled the leading Chinese PV firms to continue their aggressive investment strategies as company balance sheets around the world began suffering from the falling module prices. After a slight dip in investment activity in 2009, Chinese firms were able to regain the previous levels of public equity investment in 2010, mobilizing approximately six times the amount of public equity compared to Germany[9] [F6].

These supply-side policies went hand in hand with an important shift towards more active deployment support, which had remained minimal until 2009. As market uncertainty in Germany and other major European markets increased, China's central government finally introduced the first significant measures to promote a domestic market, including the Golden Sun programme launched in 2009 (Martinot, 2010; SEMI and CPIA, 2011). Results began emerging in the subsequent years. The market grew by over 300% in 2010 and 500% in 2011, reaching 2,5 GW of annual installed capacity (IEA-PVPS, 2012) [F5].

Finally, with the increasing size and maturity of the Chinese PV industry, its initial focus on acquiring technological capacities shifted towards a more pro-active role in the development of technology. Four Chinese firms had entered the top ten manufacturers in terms of absolute R&D spending in 2009 (Breyer et al., 2010, p.9–10). Increasingly formalized partnerships with UNSW and other international R&D centres[10] helped firms to integrate cutting-edge research into their production lines via licensing and collaborative R&D, while emerging as a new driver of knowledge development for the same R&D centres (Bullis, 2011; Quitzow, 2013) [F2/3]. Public R&D spending had not emerged as a major impetus for "knowledge development" (Huo and Zhang, 2012). Private R&D efforts were almost exclusively aimed at cost reductions and cell efficiency improvements in an effort to build market share [F4] (For a summary of key functional dynamics, see Table 4).

[9] Based on global investment data from Bloomberg New Energy Finance Investment Database.
[10] For the most part, Germany's major research institutes did not participate in these developments, remaining committed to their German counterparts.

5.3.3. Synthesis: motors of innovation and cross-country inter-dependencies

The overall growth of the period was enabled by a combination of the resilient demand-side policy framework in Germany, on the one hand, and the continued expansion of manufacturing capacity in China. In Germany, the significant over-achievement of deployment targets began introducing a degree of uncertainty about future policy developments. For the time being, however, Germany did not make any fundamental policy adjustments. This fuelled rapid investments in PV systems to capture sharply increased profit margins before more drastic policy changes were introduced. Germany's motor of innovation is, therefore, labelled "Rent-Driven Market Acceleration." Simultaneously, governance of China's PV sector enabled further expansion of manufacturing capacity, despite clear signals that this would most likely exacerbate the emerging over-supply in the sector. These aggressive investment strategies were aimed at increasing market share and benefited from the continued mobilization of international financial resources as well as increasing central government support (Zhang and White, 2012). The corresponding motor of innovation is labelled "Resource Mobilization for Expansion of Market Share."

Most existing transnational linkages were expanded or deepened during this period, increasingly taking the shape of traditional supply chain arrangements. Commercial linkages between Chinese module producers and German players both in the equipment industry and the areas of project development and distribution expanded in tandem with industry growth. Moreover, R&D partnerships with leading research centres were further institutionalized. Other linkages, such as informal coaching by German manufacturers or OEM agreements involving major Chinese module manufactures, were discontinued.

This increasingly mature global production network emerged as the backbone of a co-evolutionary process of cumulative causation involving China and Germany. In a first step, Chinese aggressive investment in low-cost module manufacturing represented the key to rapidly declining module prices and – via the commercial linkages outlined above – to the exponential growth of the German PV market. While there is a considerable degree of dispute regarding the question of how Chinese firms have been able to sustain these low prices, a number of contributing factors may be cited. Firstly, Chinese firms have focused almost exclusively on the low-cost production of silicon-based solar modules and have exploited China's enabling environment for low-cost manufacturing. Semi-automated production processes and the increasing availability of domestically produced low-cost equipment and components have helped reduce costs in China compared to other countries (Davila et al., 2009; Gallagher and Zhang, 2013; Suntech Power Holdings, 2005; Yingli Green Energy Holding, 2007). Secondly, the size advantage of Chinese firms enabled significant economies of scale, which translated into an estimated Chinese cost advantage (vis-à-vis Europe or the US) of approximately $0,12 per Wp (Goodrich et al., 2011). Thirdly, Haley and Haley (2013) have argued that investment support measures, subsidies and access to abundant financial resources have provided Chinese manufacturing firms with

Table 5
Motors of innovation and functional patterns in Germany and China and cross-country inter-dependencies (2009–2010/2011).

Motors of innovation and functional patterns	
Germany	China
Rent-driven market acceleration	Resource mobilization for expansion of market share
Transnational linkages	
• Further decline in importance of linkages between Japanese and American module producers and German distributors and project developers [TL1] • Discontinuation of informal coaching of certain Chinese pioneers by German manufacturers [TL6] • Expansion of commercial linkages between Chinese firms and German manufacturers, distributors and project developers via supply, distribution and OEM agreements (with a particular growth in cell and wafer supply agreements) [TL4] • Expansion of all other linkages in tandem with industry growth	
Indirect spillovers and feedbacks	
• Feedbacks: Growing market share of Chinese industry contributes to policy uncertainty in Germany [F2] ⇒ Growing market uncertainty in Germany (and Spain) leads to the introduction of ambitious deployment policies in China [F3]	

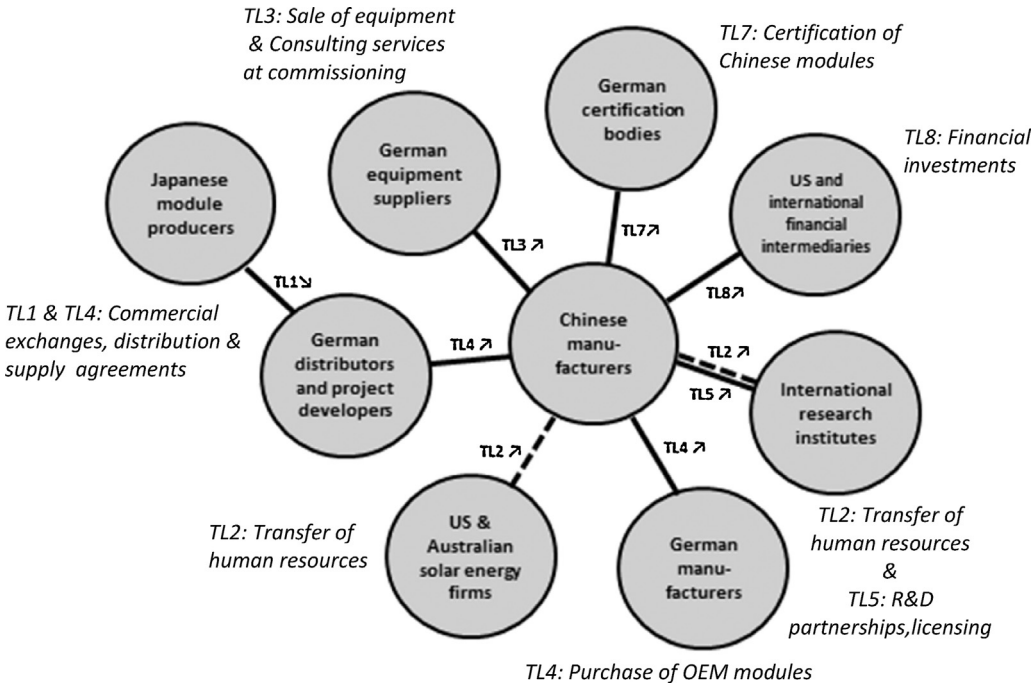


Fig. 6. Growth and adjustment of transnational linkages (2009–2010/2011).

an (unfair) cost advantage. Finally and most importantly, Chinese investments created an unprecedented over-supply, resulting in strongly price competitive firm strategies focused on increasing and defending market share (Mints, 2012; Willeke and Räuber, 2012) (Table 5) (Fig. 6).

In a second step, these market and industry dynamics led to increased uncertainty in the German market. Next to concerns regarding the growing cost of deployment support, the rise of Chinese firms raised important questions regarding desired industrial policy gains [FB2]. In a third step, these growing market uncertainties – further compounded by the collapse of the Spanish market in 2009 – triggered important policy feedbacks in China. Faced with a large-scale PV industry, the central government not only declared the sector a "strategic emerging industry," but it also rapidly ramped up domestic deployment support [FB3].

6. Discussion of key findings and questions for further research

The above analysis has helped unpack the co-evolutionary process of TIS development in solar PV technologies during the period from 1999/2000 to 2010/2011. It revises the notion of a lead market, where an innovation reaches significant market penetration before diffusing to other lag markets (Beise, 2004). Rather it gives rise to a more nuanced understanding of how country-level asymmetries shape the development and geographic reconfiguration of emerging industries. The confluence of German demand-side dynamics and the country's strength in the export of production equipment, on the one hand, and China's ability to reduce manufacturing costs and exploit international knowledge partnerships and financial markets, on the other, contributed to a co-evolutionary process of cumulative causation.

These findings highlight the importance of considering how TIS functional dynamics are embedded in distinct yet co-evolving national (or other spatially defined) systems of innovation and production. Traditionally, co-evolutionary approaches in the literature on innovation and technical change have focused on the co-evolution of technology, industry and institutions (Nelson, 1995). Approaches from

transitions studies have broadened this to incorporate for instance the role ecosystems or user practices (Foxon, 2011). Economic geographers have pointed to the co-evolution of industries, firms and networks (Terwal and Boschma, 2011) as well as industries and regional clusters (Schamp, 2010). This study adds a further dimension to this discussion, demonstrating how the global TIS in the solar PV sector was shaped by the dynamic co-evolution of two distinct, spatially defined innovation systems.

Moreover, the study has highlighted that not only transnational linkages or networks but also more indirect spillovers and feedbacks play a key role in understanding this co-evolutionary process. By differentiating these three types of cross-country influences, this paper has taken an important step towards a more nuanced conceptualization of transnational influences in TIS development. Building on this, future research might explore different types of linkages, spillovers and feedbacks and how they relate to both functional dynamics and structural features in the different country-level TIS.

This includes a further exploration of how transnational linkages may influence dynamics within individual TIS functions, as proposed by Gosens et al. (2013) and Binz et al. (2014). An equally relevant question is how specific structural features, i.e. actors and institutions, are able to establish or facilitate transnational linkages across multiple system functions and thus act as pivotal nodes within a global TIS. While this role is frequently played by so-called lead firms in more established industries (Ernst, 2002), this study suggests that so-called transnational entrepreneurs (Drori et al., 2009) may play a vital role in this regard in emerging TIS, although an important degree of variation may be expected across industries and technology fields.

In addition, exploring the specific roles and types of cross-country feedbacks in TIS development represents an important area for further research. Among other things, this case study raises the question whether cross-country feedbacks are mainly a phenomenon of emerging, policy-driven markets or whether similar feedbacks occur in conventional markets. The preceding analysis has revealed that cross-country feedbacks in the PV sector have been closely related to the realm of policy. Firstly, the exponential market growth in Germany and Spain was not only enabled by rapid price declines. Equally important was the initial rigidity of the feed-in tariff schemes, generating windfall profits for investors. Secondly, the market uncertainty that followed was closely correlated to the European policy response, which in turn motivated policy makers in China to launch domestic deployment policies. In other words, policy developments were both an important cause and a consequence of the cross-country feedbacks discussed in this article. Although cross-country feedbacks are also present in the broader economy – as witnessed in multiple financial crises over the past decades – these dynamics are cross-sectoral in nature and largely mediated by international financial markets. A working hypothesis might, therefore, be that in the absence of significant policy intervention, intra-sectoral feedbacks are likely to be less severe.

In a similar vein, this raises the question whether the pattern of spatial development observed in the PV sector – i.e. the relocation of industry followed by a relocation of the market – also represents a particularity of policy-driven TIS dynamics. As this case study has demonstrated, not only were policy developments an important driver of the market, but industrial developments have also played a key role in shaping market support policies. Most importantly, Chinese industry growth gave rise, via the feedbacks described above, to the country's demand-side policies. As industry structure has become even more concentrated in Asia, this trend has continued with China emerging as the largest market for new installations in 2013. In Germany, on the other hand, reductions in manufacturing capacity have coincided with weakening policy support. The result is an anticipated drop in market demand of approximately 50% for 2013.

Finally, an important empirical finding relates to the role that cost reduction and the development of scale economies have played in the development of the global TIS for solar PV. The exponential growth witnessed in the period from 2008 to 2010/2011 was strongly dependent on the sudden decline in module prices. The case study reveals that the underlying increase in manufacturing capacity and the related process innovations were not simply a by-product of market expansion or factor price-driven relocation of production. Rather they emerged as the result of Chinese firm strategies focused on low-cost manufacturing and the rapid development of scale economies. These were supported by a combination of China's selection environment for hi-tech entrepreneurship and the

successful exploitation of transnational linkages. This differs distinctly from the dynamics in Germany, where firms were both less concerned with price-based competition and more hesitant to scale up manufacturing. It might thus be argued that the process of cost reduction represents an additional system process, which merits attention in the analysis of TIS functional dynamics, especially when considering spatial aspects of TIS development. The ability (and willingness) to scale-up production and reduce unit costs represented a key driver of the co-evolutionary process of spatial reconfiguration in the PV sector.

7. Implications for policy assessment

Unpacking the process of spatial reconfiguration in this way also offers important entry points for a more nuanced assessment of policies in the solar energy sector. On the one hand, weakening policy support for solar energy and the consequent reduction in market demand in Germany demonstrate that dynamic feedbacks across country-level TIS may have negative implications for domestic TIS development. On the other hand, the process of cumulative causation unleashed by Germany's feed-in tariff is continuing in China and a range of other countries. Hence an assessment of Germany's solar energy policy over the past decade should also take into consideration these international dynamics and their broader implications for the advancement of renewable energy technologies and climate policy objectives. Unfortunately, many German assessments have focused narrowly on country-level industrial policy objectives. An increasingly popular notion is that Chinese supply-side subsidies have undermined German industrial leadership in the sector. This article provides ample evidence that this is clearly an over-simplified reading of developments. Rather, industry growth and cost reductions were driven by resource mobilization both on the demand- and on the supply-side, i.e. in Germany and in China, respectively. While Germany's feed-in tariff played a crucial role in unleashing demand for solar PV systems, alone it was not able to drive the observed growth of the TIS. Equally important were transnational entrepreneurial strategies in China.

In the same vein, a one-sided focus on demand-side policies does not provide a sufficient basis for generating the competitive advantages anticipated by German policy makers. Germany's ambitious deployment policies have to be seen in conjunction with supply-side factors, like its collaborative R&D landscape, its strength in equipment manufacturing and its traditional export-orientation. These characteristics play an important role in explaining the continued strength of its suppliers of PV equipment (see also Dewald & Fromhold-Eisebith, 2015). Simultaneously, they represented a key factor in enabling the rapid rise of Chinese module suppliers. In the present-day context of globalized value chains, lead market development cannot be separated from these processes of specialization and supply chain segmentation. Hence, evaluating the success of deployment policies in generating competitive advantage must take a more nuanced view, differentiating between different segments of the supply chain and their inter-relationships.

Of course, the described development trajectory represents only one possible pattern of spatial reconfiguration (Quitzow et al., 2014). Future research might, therefore, investigate different patterns of spatial reconfiguration and how these are related to factors, like the role and design of policy in the sector, technology characteristics or forms of value chain governance. This would provide important additional insights for evaluating the link between deployment policies and competitive advantage.

Acknowledgments

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Appendix. Overview of interview partners

The following list provides an overview of interviews conducted in China and Germany in the period from September 2012 to June 2013. Due to the greater availability of publically available data and secondary sources on the development of the PV sector in Germany, a larger number of interviews were conducted in China.

Interview partners – China

- First tier manufacturing firms (4 interviews)
- Second tier manufacturing firms (1 interview)
- Project developer (1 interview)
- Local government officials in Jiangxi Province and Beijing (2 interviews)
- Representatives from the School of Photovoltaic Engineering, Nanchang University, Jiangxi Province (1 interview)
- Chinese Renewable Energy Industries Association (1 interview)
- China Photovoltaic Industry Association (1 interview)
- Ministry of Industry and Information Technology (1 interview)
- National Energy Administration (1 interview)
- China National Renewable Energy Centre (1 interview)
- German Agency for International Cooperation (GIZ)–Beijing Office (1 interview)
- Europe-China Clean Energy Centre (1 interview)
- IEA PVPS (1 interview)
- University experts (4 interviews)

Interview partners–Germany

- Manufacturing firms (3 interviews)
- Equipment producer (1 interview)
- Project developer (1 interview)
- Expert in renewable energy financing (1 interview)
- Representative of Solar Valley Mitteldeutschland (1 interview)
- Sector experts/consultants (3 interviews)
- PV research centre (2 interviews)

Interview partners–International

- International research centres (2 interviews)

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